



# Computer Graphics



## Chapter 1: What is Computer Graphics?

*Computer Graphics = Using a computer to create, store, and show pictures on a screen.*

Think of it like this: when you draw with crayons on paper, that is art. When you do the SAME drawing using a computer, that is Computer Graphics!



**Real Life**

Video games like Minecraft, Fortnite — all made with computer graphics!



### Where is it used?

- Entertainment — movies, cartoons, games
- Design — architects draw buildings using it
- Medical — doctors see body scans (X-ray, MRI)
- Education — learning apps and simulations
- Science — scientists show data as colorful charts



### Think Questions

Q1

Can you name 2 video games that use computer graphics?

Q2

Why do you think doctors use computer graphics to look inside the body?

Q3

What is the difference between drawing on paper and drawing on a computer?



## Chapter 2: Interactive vs Passive Graphics

There are TWO types of computer graphics. Think of it like this:



**Interactive**

YOU can change what happens! Like moving your character in a video game.



**Passive**

You just WATCH. Like watching a cartoon on TV. You cannot change it.

### Interactive Graphics — You control it!

- Video Games — you move the character
- CAD Software — engineers change 3D designs
- Virtual Reality (VR) — you look around in a fake world
- Data Dashboards — click to filter a chart

## Passive Graphics — Just watch!

- Movies and TV shows
- Printed posters and books
- Pre-made animations for ads

### Think Questions

- |    |                                                       |
|----|-------------------------------------------------------|
| Q1 | Is YouTube interactive or passive? Why?               |
| Q2 | Name one app on a phone that is interactive graphics. |
| Q3 | What makes a video game DIFFERENT from a movie?       |

## Chapter 3: Display Systems & The Display Processor

*A Display Processor = the brain that converts computer data into pictures you see on screen.*

Imagine your computer has a special helper inside. Its job is to take numbers/instructions and turn them into the pretty images on your monitor.

### 4 Parts of the Display Processor

- **1. Display File Memory**

Stores the list of instructions for drawing the picture.

- **2. Display Controller**

Makes sure everything happens at the right time (like a traffic light).

- **3. Display Generator**

Actually draws the lines, curves, and characters.

- **4. Display Console**

The screen you see (like CRT, keyboard, light pen).



#### Example

When you open Paint and draw a line, the Display Processor takes your mouse movement and converts it into pixels on screen.

### Think Questions

- |    |                                                                    |
|----|--------------------------------------------------------------------|
| Q1 | What does the Display File Memory store?                           |
| Q2 | Why do you think timing is important for showing images correctly? |
| Q3 | Which part of the display processor actually draws the shapes?     |

## Chapter 4: Raster Scan vs Random Scan

### Raster Scan — like mowing a lawn

*The screen is drawn line by line, left to right, top to bottom — like how you read a book!*

Each line is called a SCAN LINE. The computer fills every tiny dot (called a PIXEL) on each line.

- Used in: TVs, computer monitors, phone screens
- Makes grid of pixels
- Very common today

### Random Scan — like an Etch-a-Sketch

*The beam only goes to specific points to draw lines — not the whole screen. Like using a pen to connect dots!*

- Draws only lines and shapes (no filled colors)
- Old technology — used in vector displays
- Example: Pen plotters



#### Analogy

Raster Scan = coloring every square in a grid. Random Scan = connecting specific dots to make shapes.

### Think Questions

- Q1 Your phone screen — is it raster scan or random scan?
- Q2 Why do you think raster scan is better for showing photos?
- Q3 What is a PIXEL? Can you see pixels if you zoom into a photo on your phone?

## Chapter 5: CRT — The Old TV Screen

*CRT = Cathode Ray Tube. It is an OLD technology used in big, heavy monitors and TVs before flat screens.*

Remember those old bulky TVs that looked like a box? That was a CRT!

### How CRT Works — Step by Step

1. Electron Gun fires electrons (like tiny invisible bullets) from the back.
2. Electrons hit the phosphor coating on the screen's front glass.
3. Phosphor GLOWS where the electrons hit — making a dot of light!
4. Millions of these glowing dots form the picture.
5. 3 colors used: Red + Green + Blue (RGB) combine to make all colors.

## 5 Key Parts of a CRT

- Electron Gun — shoots electrons
- Control Electrode — turns beam ON/OFF
- Focusing System — keeps the beam sharp (like focusing a torch)
- Deflection Yoke — steers the beam up/down/left/right
- Phosphor Screen — glows when electrons hit it



### Fun Fact

Your phone screen shows millions of pixels using tiny colored lights (LEDs) — similar idea, but smaller and flat!



### Think Questions

- Q1 What are the 3 colors used in CRT to make all other colors?
- Q2 Why do you think old CRT TVs were so big and heavy compared to today's flat screens?
- Q3 What happens when electrons hit the phosphor coating?



## Chapter 6: Display Devices

Many types of screens exist. Here is a simple comparison:

| Display Type             | How It Works                              | Example                                  |
|--------------------------|-------------------------------------------|------------------------------------------|
| CRT                      | Electron beam hits phosphor               | <i>Old bulky TV</i>                      |
| LCD (Flat Panel)         | Liquid crystals block/pass light          | <i>Laptop screen</i>                     |
| OLED                     | Organic compounds emit light directly     | <i>iPhone screen</i>                     |
| Lookup Table (LUT)       | Maps color numbers to real display colors | <i>Color correction in video editing</i> |
| Direct View Storage Tube | Phosphor holds image without refresh      | <i>Old engineering displays</i>          |



### Think Questions

- Q1 Which display type is your phone most likely using — CRT or OLED?
- Q2 What advantage does a flat panel (LCD/OLED) have over an old CRT?
- Q3 What does a Lookup Table (LUT) do with colors?

## Chapter 7: 2D Transformations

*Transformations = Moving, resizing, or flipping objects on the screen.*

Imagine you have a drawing of a car. Transformations let you:

### 1. Translation — Moving

Moving an object from one place to another. The shape stays the SAME — only its position changes.



#### Example

In a game, when your car drives forward — that is Translation! Position changes, car looks the same.

### 2. Scaling — Resizing

Making an object BIGGER or SMALLER. The shape stays the same, but the size changes.



#### Example

When you pinch your fingers on a photo to zoom in — that is Scaling!

### 3. Rotation — Spinning

Rotating means turning the object around a point (called the pivot point).

- Positive angle = Counterclockwise (anti-clockwise)
- Negative angle = Clockwise



#### Example

Spinning a wheel in a car game = Rotation. Every point of the wheel turns by the same angle.

### 4. Reflection — Mirror Image

Creating a mirror image of an object. It is like flipping it.

- Reflect about X-axis = flip UP/DOWN
- Reflect about Y-axis = flip LEFT/RIGHT
- Reflect about  $y=x$  line = flip diagonally



#### Example

A butterfly's wings are reflections of each other!

### 5. Shearing — Slanting

Shearing DISTORTS the shape — it shifts one part while the rest stays. Like pushing the top of a rectangle sideways to make a parallelogram.



#### Example

Italic text in Microsoft Word is a form of shearing — letters are slanted!

## Think Questions

- |    |                                                                                                    |
|----|----------------------------------------------------------------------------------------------------|
| Q1 | If you move a circle from the left side of the screen to the right — which transformation is that? |
| Q2 | What transformation makes a small image bigger (like zooming in)?                                  |
| Q3 | If you flip a photo upside down — which transformation did you use?                                |
| Q4 | Can you think of a real-life example of Rotation in a computer game?                               |
| Q5 | What is the difference between Rotation and Reflection?                                            |

## Chapter 8: Window, Zooming & Panning

### Window

*A Window = a rectangular area on the screen where your graphics are shown.*

Just like a window in your house lets you see only PART of the outside world, a Graphics Window shows you only part of the virtual scene.

- Has its own Coordinate System (X and Y axes)
- Has a Resolution — more pixels = sharper image
- Has a Viewport — the actual visible part of the window
- You can resize, minimize, or close it

### Zooming

Zooming changes how big the image appears on screen by replicating or reducing pixels.

- Zoom In = pixels get bigger, image looks closer
- Zoom Out = pixels get smaller, more area visible



#### Example

In Google Maps, scrolling down zooms in to see street details. Scrolling up zooms out to see the whole country.

### Panning

Panning = moving the view across the scene without zooming. Like sliding a big picture left or right to see different parts.



#### Example

Dragging the map in Google Maps to move to a different city — that is Panning!

## Think Questions

- |    |                                                                                |
|----|--------------------------------------------------------------------------------|
| Q1 | What is the difference between Zooming and Panning?                            |
| Q2 | When you drag a map on your phone to see another area, what technique is that? |

Q3 What is a Viewport? How is it different from a Window?

## Chapter 9: Clipping

*Clipping = Only draw what is INSIDE the view. Cut off (clip) what is OUTSIDE.*

Imagine you have a big painting but only a small window to look through. You only see the part of the painting inside the window frame — the rest is CLIPPED (hidden).

### Why is Clipping useful?

- Saves computer resources — no need to draw things you cannot see
- Lets you crop or mask parts of an image
- Used for 3D objects — hides back faces or off-screen objects
- Used for copying/moving only a selected region



#### Example

In photo editing, when you "crop" a photo — you are clipping! You keep only the part inside the crop rectangle.



### Think Questions

- Q1 Why would a game not draw objects that are behind you (outside the view)?
- Q2 Is cropping a photo the same as clipping? Why?
- Q3 What would happen if a computer drew everything — even things outside the screen?



## Chapter 10: Pointing & Positioning Devices

*These are input tools that let you point to or place items on the screen.*

### 1. Mouse

- Move cursor on screen to point and click
- Used in Photoshop, CAD software, web browsing



#### Daily Life

Every time you click a button on a website, you use mouse pointing!

### 2. Light Pen

An older tool shaped like a pen. You touch the screen with it, and it detects where light is on the display at that point.

- Has a light-sensing tip
- Detects screen position using light timing
- Used in very early computers

### 3. Joystick

- Used in gaming and 3D navigation
- VR controllers use joystick-like movements
- Also used in CAD and engineering software

### Think Questions

- |    |                                                                 |
|----|-----------------------------------------------------------------|
| Q1 | What input device do you use most to interact with a computer?  |
| Q2 | How does a Light Pen know WHERE on the screen you are pointing? |
| Q3 | Why would a joystick be useful in a 3D game?                    |

## Chapter 11: Rubber Band & Dragging

### Rubber Band (Elastic) Technique

*When you draw a line and can see it stretch from your starting point to your cursor — that is the Rubber Band technique!*

Named because it looks like a rubber band being stretched between two points. As you move your mouse, the line 'stretches' with you.



#### Example

In MS Paint, when you draw a line — the preview line that follows your cursor before you click is the Rubber Band technique.

### Dragging

Dragging = holding the mouse button down and moving an object to a new position.

6. Click and hold an object.
7. Move the mouse — object follows the cursor.
8. Release the button — object stays at new position.



#### Example

Moving a file from one folder to another on your computer desktop — that is Dragging!

### Think Questions

- |    |                                                                      |
|----|----------------------------------------------------------------------|
| Q1 | Can you think of a time you used Dragging on a computer or phone?    |
| Q2 | What is the difference between Rubber Band technique and Dragging?   |
| Q3 | In MS Paint, try drawing a line — do you see the rubber band effect? |



## Chapter 12: Shading — Making 3D Look Real

*Shading = Adding shadows and light effects to make flat shapes look 3D and realistic.*

Imagine drawing a ball. Without shading it looks like a flat circle. With shading — it looks ROUND!

### Types of Lighting

- Ambient Light — everywhere background light (like a cloudy day — no strong shadow)
- Diffuse Light — rough surface, scatters light in all directions (matte finish)
- Specular Light — shiny surface, reflects light in one direction (the bright spot on a shiny ball)

### Shading Models

#### Flat Shading

Each face/polygon gets ONE flat color based on light angle. Fast but looks blocky.



##### Example

Low-polygon game characters from old PS1 games — each flat triangle had one solid color.

#### Gouraud Shading (1971)

Calculates color at each CORNER (vertex) of a polygon, then smoothly blends across the surface. Smoother than flat, but misses small highlights.



##### Example

A basketball in a game looks rounder with Gouraud shading — colors blend from dark to light.

#### Phong Shading (1975)

Calculates color at EVERY PIXEL. Very realistic. Shows specular highlights accurately. Used in modern games.



##### Example

The shiny reflection on a car hood in a racing game — that is Phong shading at work!

### Advanced Shading Techniques

- Texture Mapping — wrapping an image (texture) around a 3D object
- Bump Mapping — faking bumps/wrinkles without adding geometry

- Shadow Mapping — calculating which areas are in shadow
- Ray Tracing — tracing light rays for ultra-realistic images
- Global Illumination — simulating bounced light (light reflecting off walls)
- Subsurface Scattering — light passing through skin/milk/marble
- PBR (Physically Based Rendering) — most realistic; used in modern games

#### Real Game Example

Games like Cyberpunk 2077 use Ray Tracing for reflections and shadows that look like real photos!

## Surface Properties that Affect Shading

- Reflectivity — how much light bounces back (mirror vs chalk)
- Transparency — does light pass through? (glass vs wood)
- Color — what wavelengths does the surface reflect?
- Texture — rough vs smooth surface
- Normal Vector — which direction is the surface facing?

## Think Questions

- Q1 Why does a ball drawn without shading look flat?
- Q2 What is the difference between Gouraud and Phong shading?
- Q3 Which shading technique would make a character's skin look most realistic?
- Q4 What does Texture Mapping do to a 3D object?
- Q5 Can you think of a real object that uses Subsurface Scattering? (Hint: look at your hand in sunlight)

## Quick Recap — All Topics at a Glance

| Topic                | Simple Meaning                           | Real Life Example                   |
|----------------------|------------------------------------------|-------------------------------------|
| Computer Graphics    | Making/showing pictures on a computer    | <i>Video games, animated movies</i> |
| Interactive Graphics | You control and change it                | <i>Playing a video game</i>         |
| Passive Graphics     | You just watch                           | <i>Watching a cartoon</i>           |
| Raster Scan          | Draw screen line by line, pixel by pixel | <i>Your phone/TV screen</i>         |
| Random Scan          | Draw only specific lines                 | <i>Old vector monitors</i>          |
| CRT                  | Old screen using electron beam           | <i>Old bulky TV</i>                 |

|                        |                                     |                                 |
|------------------------|-------------------------------------|---------------------------------|
| <b>Translation</b>     | Moving an object                    | <i>Driving a car left/right</i> |
| <b>Scaling</b>         | Resizing an object                  | <i>Zooming into a photo</i>     |
| <b>Rotation</b>        | Spinning an object                  | <i>Spinning a wheel</i>         |
| <b>Reflection</b>      | Mirror flip of an object            | <i>Butterfly wings</i>          |
| <b>Shearing</b>        | Slanting/distorting an object       | <i>Italic text</i>              |
| <b>Window</b>          | Visible area of the screen          | <i>App window on desktop</i>    |
| <b>Zooming</b>         | Making view bigger/smaller          | <i>Google Maps zoom</i>         |
| <b>Panning</b>         | Sliding view across scene           | <i>Dragging Google Maps</i>     |
| <b>Clipping</b>        | Only draw what is inside view       | <i>Photo cropping</i>           |
| <b>Rubber Band</b>     | Stretchy line preview while drawing | <i>Line preview in MS Paint</i> |
| <b>Dragging</b>        | Moving an object with click & hold  | <i>Moving files on desktop</i>  |
| <b>Shading</b>         | Adding light/shadow to look 3D      | <i>Ball that looks round</i>    |
| <b>Gouraud Shading</b> | Smooth color blend per vertex       | <i>Smooth 3D characters</i>     |
| <b>Phong Shading</b>   | Per-pixel shading, very realistic   | <i>Shiny car hood in games</i>  |

